

# Test Case Specification Document

## InterviewAce — Frontend Team

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|---------------|----------------------------------------|
| Project       | InterviewAce (Job Interview Prep Tool) |
| Document Type | Test Case Specification                |
| Team Scope    | Frontend (React / Vite)                |
| Prepared By   | Frontend Team                          |
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### 1. Scope & Objectives

This document defines the frontend test cases for InterviewAce. The platform enables job seekers to select a role, practice interview questions, submit answers, receive AI-generated scores across four axes (Technical Depth, Clarity, Completeness, Structure), and track their progress over time. Test cases focus exclusively on the **frontend layer** — rendering, state management, user interactions, validation, API integration handoffs, and responsive design. Backend scoring logic, JWT authentication, and the AI/ML pipeline are out of scope.

### 2. Test Cases

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| TC-FE-01                         | Job Role Selection — All Tracks Render Correctly                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PASS                              |
| Use Case: UC-01: Select Job Role |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Priority: High   Type: Functional |
| Preconditions                    | User is on the landing/home page. No role or level previously selected.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |
| Test Steps                       | <ul style="list-style-type: none"><li>• Navigate to the home page (/).</li><li>• Observe the role-selection UI (cards or dropdown).</li><li>• Click on the 'Software Engineer (SWE)' role.</li><li>• Verify the page/state updates to reflect the role selection.</li><li>• Repeat steps 3–4 for 'Data Science', 'PM', and 'Behavioral' tracks.</li><li>• For each role, select an experience level: 'Intern' and 'New Grad'.</li><li>• Verify the level selection is reflected in the UI state.</li></ul>                    |                                   |
| Expected Results                 | <ul style="list-style-type: none"><li>• All four role options (SWE, Data Science, PM, Behavioral) are visible and correctly labeled.</li><li>• Both experience levels (Intern, New Grad) are visible and selectable for every role.</li><li>• Selecting a role AND a level highlights each choice and enables the 'Start Practice' button.</li><li>• The selected role and level are stored in component state (or local context) and reflected in the UI.</li><li>• No console errors are thrown during selection.</li></ul> |                                   |
| Notes / Risk                     | Verify responsiveness — role cards and level selectors must stack vertically on mobile viewports (<768 px).                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |

| TC-FE-02 Practice Interview — Question Display & Navigation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PASS                              |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Use Case: UC-02: Practice Interview                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Priority: High   Type: Functional |
| Preconditions                                               | User has selected 'SWE' role and an experience level. User is on the practice session screen.                                                                                                                                                                                                                                                                                                                                                                            |                                   |
| Test Steps                                                  | <ul style="list-style-type: none"> <li>• Enter the practice session after selecting the SWE role.</li> <li>• Observe the first interview question rendered on screen.</li> <li>• Confirm a text-input area (textarea) is present.</li> <li>• Click 'Next Question' or equivalent navigation control.</li> <li>• Verify the second question loads without a full page reload.</li> <li>• Click 'Previous Question' (if available) and confirm navigation back.</li> </ul> |                                   |
| Expected Results                                            | <ul style="list-style-type: none"> <li>• At least one curated question is displayed for the SWE role.</li> <li>• A textarea / answer input is visible and accepts keyboard input.</li> <li>• Navigation between questions works; the question counter updates (e.g., '2 / 10').</li> <li>• Previously typed answers are preserved when navigating back.</li> <li>• A progress indicator (e.g., progress bar) reflects question advancement.</li> </ul>                   |                                   |
| Notes / Risk                                                | Test with all four role tracks. Edge case: empty question bank should show a friendly error.                                                                                                                                                                                                                                                                                                                                                                             |                                   |

  

| TC-FE-03 Answer Submission — API Call & Loading State |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PASS                               |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Use Case: UC-03: Submit Answer                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Priority: High   Type: Integration |
| Preconditions                                         | User is on a question screen with a non-empty answer typed in the textarea.                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                    |
| Test Steps                                            | <ul style="list-style-type: none"> <li>• Type a valid answer (≥20 characters) into the answer textarea.</li> <li>• Click the 'Submit' button.</li> <li>• Observe the UI immediately after clicking.</li> <li>• Wait for the API response (mocked or live).</li> <li>• Observe the resulting state once the response is received.</li> </ul>                                                                                                                                                                          |                                    |
| Expected Results                                      | <ul style="list-style-type: none"> <li>• Submit button becomes disabled immediately after click (prevents double-submission).</li> <li>• A loading spinner or skeleton placeholder is shown while awaiting AI scoring.</li> <li>• On success, the user is navigated to (or the section expands to show) the score &amp; feedback view.</li> <li>• On API error (5xx), a user-friendly error toast/banner is displayed.</li> <li>• The answer text is not cleared until a successful response is received.</li> </ul> |                                    |
| Notes / Risk                                          | Use network throttling (Slow 3G) in DevTools to verify the loading state visibility.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                    |

  

| TC-FE-04 Score & Feedback Display — 4-Axis Horizontal Bar Chart |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PASS                              |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Use Case: UC-04: View Score & Feedback                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Priority: High   Type: Functional |
| Preconditions                                                   | A valid answer has been submitted and the backend has returned a score object with all four axes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                   |
| Test Steps                                                      | <ul style="list-style-type: none"> <li>• Submit a complete answer for any role question.</li> <li>• Navigate to (or observe the automatic transition to) the feedback screen.</li> <li>• Observe the horizontal bar chart rendered with four axes.</li> <li>• Verify each axis label is correct.</li> <li>• Observe the textual feedback section below the chart.</li> <li>• Read the actionable improvement tips listed.</li> </ul>                                                                                                                                          |                                   |
| Expected Results                                                | <ul style="list-style-type: none"> <li>• A horizontal bar chart (Recharts) with exactly four labeled axes is rendered: Technical Depth, Clarity, Completeness, Structure.</li> <li>• Each bar displays a numeric score (e.g., 0–100) matching the API response.</li> <li>• The chart is interactive or clearly annotated — hovering a bar shows the score.</li> <li>• The feedback section lists at least one improvement tip per missing rubric element.</li> <li>• All text in the feedback section is readable (min font size 12px, sufficient color contrast).</li> </ul> |                                   |
| Notes / Risk                                                    | Accessibility: chart must include ARIA labels or a textual score summary for screen readers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                   |

| TC-FE-05 Progress Tracking — Answer History & Score Trends |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PASS                                |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Use Case: UC-05: Track Progress                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Priority: Medium   Type: Functional |
| Preconditions                                              | User has submitted answers for at least 3 different questions across different sessions.                                                                                                                                                                                                                                                                                                                                                                                      |                                     |
| Test Steps                                                 | <ul style="list-style-type: none"> <li>• Navigate to the 'Progress' or 'History' section of the app.</li> <li>• Observe the answer history list.</li> <li>• Click on a past answer entry to expand or view its details.</li> <li>• Observe the score trend chart (line/bar chart over time).</li> <li>• Filter or sort the history (if controls exist).</li> </ul>                                                                                                            |                                     |
| Expected Results                                           | <ul style="list-style-type: none"> <li>• All previously submitted answers are listed in reverse-chronological order.</li> <li>• Each history entry displays: question snippet, role, date, and overall score.</li> <li>• Clicking an entry expands it to show the full answer and 4-axis scores.</li> <li>• A score trend chart shows improvement (or decline) over time per axis.</li> <li>• Filter/sort controls (if present) correctly update the visible list.</li> </ul> |                                     |
| Notes / Risk                                               | Test with 0 prior answers — empty state illustration must appear (no blank page).                                                                                                                                                                                                                                                                                                                                                                                             |                                     |

  

| TC-FE-06 Role Selection Persistence Across Page Refresh |                                                                                                                                                                                                                                                                                                                               | PASS                                            |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Use Case: UC-01: Select Job Role                        |                                                                                                                                                                                                                                                                                                                               | Priority: Medium   Type: Non-Functional (State) |
| Preconditions                                           | User has selected 'PM' role and is on the practice screen.                                                                                                                                                                                                                                                                    |                                                 |
| Test Steps                                              | <ul style="list-style-type: none"> <li>• Select the 'PM' role on the home page.</li> <li>• Start a practice session.</li> <li>• Hard-refresh the browser (Ctrl+Shift+R / Cmd+Shift+R).</li> <li>• Observe which screen loads after refresh.</li> </ul>                                                                        |                                                 |
| Expected Results                                        | <ul style="list-style-type: none"> <li>• After refresh, the user is returned to the home/role-selection screen (or the selected role is remembered if persistent state is implemented).</li> <li>• No runtime JS error appears in the console on refresh.</li> <li>• The app does not show a blank/crashed screen.</li> </ul> |                                                 |
| Notes / Risk                                            | Behavior depends on whether session/localStorage persistence is implemented.                                                                                                                                                                                                                                                  |                                                 |

  

| TC-FE-07 Empty Answer Validation — Submit Button Guard |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PASS                                         |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Use Case: UC-03: Submit Answer                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Priority: High   Type: Negative / Validation |
| Preconditions                                          | User is on the question screen with an empty textarea.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                              |
| Test Steps                                             | <ul style="list-style-type: none"> <li>• Navigate to a practice question screen.</li> <li>• Leave the answer textarea completely empty.</li> <li>• Click (or attempt to click) the 'Submit' button.</li> <li>• Type a single space character and attempt submission again.</li> <li>• Type an answer of exactly 19 characters and attempt submission.</li> </ul>                                                                                                                                                              |                                              |
| Expected Results                                       | <ul style="list-style-type: none"> <li>• The 'Submit' button is disabled when the textarea is empty.</li> <li>• An inline validation message (e.g., 'Please enter an answer') appears if the user attempts submission.</li> <li>• A whitespace-only answer also fails validation and does not trigger an API call.</li> <li>• An answer shorter than 20 characters fails validation with an appropriate message (FR-05).</li> <li>• No network request is made to the scoring endpoint for any of the above cases.</li> </ul> |                                              |
| Notes / Risk                                           | Check DevTools Network tab to confirm no outgoing request on invalid submission.                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                              |

| TC-FE-08      Responsive Layout — Mobile Viewport (375 px) |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PASS                                            |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Use Case: All Front-End Use Cases                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Priority: Medium   Type: Non-Functional (UI/UX) |
| Preconditions                                              | DevTools set to iPhone SE viewport (375 × 667 px) or physical device.                                                                                                                                                                                                                                                                                                                                                                              |                                                 |
| Test Steps                                                 | <ul style="list-style-type: none"> <li>• Open the application on a 375 px wide viewport.</li> <li>• Navigate through: Home → Role Selection (including level selector) → Practice → Score View → Progress → AI Coach.</li> <li>• Tap all interactive elements (buttons, cards, inputs).</li> <li>• Check for horizontal scrollbars or overflowing content.</li> </ul>                                                                              |                                                 |
| Expected Results                                           | <ul style="list-style-type: none"> <li>• No horizontal scroll bar appears on any screen.</li> <li>• All buttons have touch targets ≥ 44×44 px.</li> <li>• Role cards and level selectors stack vertically and are fully visible.</li> <li>• Horizontal bar chart scales proportionally and remains legible.</li> <li>• AI Coach panel renders correctly and chat input is accessible.</li> <li>• Text does not overflow its containers.</li> </ul> |                                                 |
| Notes / Risk                                               | Also test at 768 px (tablet breakpoint) for intermediate layout correctness.                                                                                                                                                                                                                                                                                                                                                                       |                                                 |

| TC-FE-09      AI Coach UI — Panel Open & Chat Interaction |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PASS                              |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Use Case: UC-06: AI Coach                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Priority: High   Type: Functional |
| Preconditions                                             | User is authenticated and has completed at least one answer submission. AI Coach feature is enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                   |
| Test Steps                                                | <ul style="list-style-type: none"> <li>• Click the 'AI Coach' button or panel trigger from any relevant screen.</li> <li>• Verify the AI Coach panel opens without a full page reload.</li> <li>• Observe the chat thread area (initially empty or with a welcome message).</li> <li>• Type a coaching question into the message input field.</li> <li>• Click 'Send' (or press Enter) and observe the UI.</li> <li>• Wait for the AI response to appear in the chat thread.</li> <li>• Send a follow-up message to verify thread continuity.</li> </ul>                                                                    |                                   |
| Expected Results                                          | <ul style="list-style-type: none"> <li>• The AI Coach panel opens and is visible without a page reload.</li> <li>• The chat thread area renders and new messages append correctly.</li> <li>• A loading indicator appears while the AI response is being fetched.</li> <li>• The AI response is appended to the thread after the loading state resolves.</li> <li>• Long messages wrap correctly and do not overflow the panel.</li> <li>• The chat panel is scrollable if the thread exceeds the panel height.</li> <li>• Closing and reopening the panel preserves the current chat thread within the session.</li> </ul> |                                   |
| Notes / Risk                                              | Verify loading state visibility under Slow 3G throttling. Test overflow/wrapping with a 500+ character AI response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |